

BUILDING LLMs FROM SCRATCH

Pipeline Architecture & Design Decisions

Why we built it this way -- tradeoffs, alternatives, and maturity assessment

Project	YouTube Playlist -> eBook Pipeline
Playlist	"Building LLMs from Scratch" (43 videos)
Goal	eBook-quality PDF, fully reproducible, zero hallucinations
Step 3 LLM	Gemini 2.5 Flash Lite (primary) + Gemini 2.5 Flash (fallback)
Step 4b LLM	Groq llama-3.1-8b-instant (reference extraction only)

1. Pipeline Overview

The pipeline converts a YouTube playlist into an eBook through six discrete, independently re-runnable steps. Each step reads from files and writes to files -- no shared in-memory state. The pipeline is fully resumable and each step can be re-run in isolation.



Already-completed steps are always skipped on re-run. Safe to interrupt and resume at any point.

2. LLM Strategy

2.1 Step 3: Gemini Only, Two Models

Chapter generation uses two Gemini models with separate daily quota pools. Flash Lite is tried first for every chapter. If it hits its daily limit (20 RPD), Flash activates automatically -- no code change, no manual intervention required.

Model	RPM	RPD	Role
gemini-2.5-flash-lite	10	20	Primary -- used for every chapter first
gemini-2.5-flash	5	20	Fallback -- separate quota, activates on 429

WHY GEMINI ONLY	Consistency. Both models receive identical system prompts, temperature=0, and the same ASCII-only instruction. Using a single provider eliminates any risk of style variation between chapters that would occur with cross-provider fallback.
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Effective daily capacity: up to 40 chapters per day (20 from each model). For a 43-video playlist this means 2 days maximum on free tier with no billing required.

2.2 Step 4b: Groq Only

Reference extraction uses Groq llama-3.1-8b-instant exclusively. This is intentionally a completely separate quota pool from step 3.

Factor	Reasoning
Separate from Gemini quota	Step 4b running on Gemini would compete with step 3 for the 20 RPD limit
8B model is sufficient	Reference extraction is pattern matching, not creative writing -- 8B handles it reliably
Low token usage	8B model uses ~10x fewer tokens than 120B, well within Groq free tier daily limits
Groq is free	No billing, no daily RPD limit equivalent to Gemini -- generous for small extraction tasks

2.3 Why Not One Model for Everything

Option	Problem
Gemini Flash Lite only	20 RPD -- cannot finish 43 chapters in one day
Gemini Flash Lite + Flash (chapters) + Gemini for refs	Refs consume chapter quota -- both tasks compete
Groq only	Style inconsistency; 120B TPD limit; cross-provider prose differences
Gemini for chapters + Groq 8B for refs (chosen)	Three separate quota pools; each task matched to right model

3. Core Design Decisions

3.1 Full Transcript -> LLM (not RAG)

Each video transcript is sent in full to the LLM to convert spoken tutorial content into written prose. This is not RAG -- no vector database, no retrieval, no embedding.

WHY	The playlist is a sequential tutorial -- each video builds on the previous. RAG retrieves by semantic similarity and destroys this order. Sending the full transcript gives the model complete context. Nothing is dropped, filtered, or de-prioritized.
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3.2 YouTube Captions vs Whisper

Factor	Whisper (rejected)	YouTube Captions (chosen)
Hardware required	GPU (Colab T4)	None -- any laptop
Speed	~15 min/video	<10s/video
Setup	Colab + pyannote + SpeechBrain	yt-dlp, one command
Code term accuracy	Same failure mode as captions	Same failure mode -- garbles identically
Reproducibility	Colab sessions expire, manual steps	Fully automated, local, scriptable

KEY INSIGHT	Both Whisper and YouTube captions fail on code terms in the same way. Whisper higher accuracy on speech does not help with the actual problem. Captions are simpler, faster, and require no hardware.
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3.3 No Vector Database

Vector DB + RAG was explicitly evaluated and rejected. The content is an ordered tutorial series -- chapter 5 assumes knowledge from chapters 1-4. RAG retrieves by similarity, not sequence, and would produce chapters that are disconnected from the pedagogical flow.

3.4 Paragraph-Based Chunking

When a transcript exceeds MAX_CHARS (30,000 characters), it is split at paragraph boundaries (double newline). It is never cut mid-sentence, mid-word, or mid-code-block.

PREVIOUS ISSUE FIXED	An earlier version hard-cut transcripts at 40,000 characters -- potentially mid-sentence. This produced garbled LLM input at chunk boundaries. Paragraph-based splitting was introduced to ensure every chunk is clean, complete text.
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4. Anti-Hallucination Design

Mechanism	What it prevents	Where implemented
temperature=0	Stochastic invention	Both Gemini models, Groq
"Only from transcript" rule	Adding content not in source video	chapter_writer.txt prompt
[Transcript unclear] flag	Silent guessing on garbled captions	chapter_writer.txt prompt
ASCII-only instruction	Unicode corruption in output	step3 prompt
Citation on every paragraph	Unverifiable claims in book	step3 prompt + PDF clickable links
References appendix	Uncited foundational paper claims	step4b_references.py
Raw transcripts saved	Full audit trail of pipeline inputs	data/clean/*.txt

5. Known Limitations

5.1 Screen-Only Code

Code written on screen without narration is absent from transcripts and therefore absent from the book. Clickable timestamp links let readers jump to the exact video moment to see what was on screen. This is disclosed in the book preface.

5.2 Caption Code Term Mangling

YouTube captions garble Python identifiers consistently: "nn.Linear" becomes "en en dot linear". The system prompt contains a reconstruction lookup table. Terms the model cannot reconstruct confidently are flagged with [Transcript unclear].

5.3 Independent Chapter Generation

Each chapter is generated without knowledge of other chapters. Automatic cross-references between chapters are not possible. This is acceptable -- each chapter being self-contained is a feature for readers who want to read selectively.

5.4 Multi-Day Generation on Free Tier

43 chapters at 40 effective RPD (20 per Gemini model) requires 2 days minimum on free tier. Already-generated chapters are never re-processed, so re-running step 3 each day resumes from where it stopped.

6. Key Decisions Log

Decision	Alternatives considered	Reason chosen
Gemini Flash Lite + Flash fallback (step 3)	Groq fallback, single model	Same provider = consistent style; two models = 40 effective RPD free
Groq 8B for step 4b	Gemini for refs, Groq 120B	Separate quota pool; 8B sufficient; very low token usage
Paragraph-based chunking	Hard char limit (40k), token-based splits	Never cuts mid-sentence; clean LLM input at every chunk boundary
temperature=0 both providers	Low temp 0.3, sampling	Determinism required for no-hallucination guarantee
YouTube captions via yt-dlp	Whisper ASR, AssemblyAI	Zero hardware; same practical accuracy on code terms; fully automated
Direct transcript -> LLM	RAG + vector DB, summarize-then-expand	Sequential tutorial must preserve order; RAG breaks this
ReportLab for PDF	WeasyPrint (GTK deps), Pandoc, LaTeX	Pure Python; no system dependencies; works on Windows without admin
File-based step outputs	In-memory pipeline, database	Enables resume across days; debug any step independently
ASCII instruction in prompt	UTF-8 throughout, post-processing only	Prevents unicode corruption at source;